

Four key SE processes are useful to introduce SE on infrastructure projects: requirements management, interface management, verification, and validation. These processes can improve infrastructure project delivery, and total life cycle view that integrates design, construction, and asset management.

4.4.6 Oil and Gas Systems

Overview of SE Applications within the Oil and Gas Domain

The emergence of SE within the Oil and Gas (O&G) domain is relatively new compared to other sectors of similar complexity. Most applications of SE have occurred within the past decade to varying levels of implementation. Due to fluctuation in oil prices, new systems with increasing complexity and efforts to reduce greenhouse gas emissions have motivated a risk-averse industry to adapt to, and in some cases drive, change. This has encouraged an entire culture known to resist change to challenge assumptions and traditional ways of working, especially working in a document-centric environment.

The greatest SE-related need has been in the system requirements definition and requirements management space. With a domain-wide focus on digitalization, the change in how requirements are defined and transmitted throughout the supply chain has benefited from an SE approach. The industry leaders have either switched, or are switching, to data-centric requirement sets. There has been collaboration between suppliers and acquirers to improve the quality and traceability of requirements and create metrics for measurement of progress. INCOSE and American Petroleum Institute (API) cooperated on some trials in 2017 and 2018 that explained the aims and elements of good requirement writing to a panel of experts involved in updating a standard. They then supported the engineers in the writing and recrafting of the content, resulting in higher quality requirements and clearly separating between instruction, information, and verification (IOGP, 2021).

Beyond requirements, additional SE practices are also being introduced in the O&G domain. For example, SE practitioners take advantage of requirement definition to develop system architectures and systematically define interfaces. By using requirement management tools, configuration management and change management of requirements can be implemented in projects. In other cases, systems thinking tools, such as context diagrams and functional trees, are used as a foundation for SE practices, such as functional modeling. Technical requirements are also leading to conversations and implementation of verification and validation strategies and realization. With the companies incorporating digital design data across disciplines and throughout the life cycle, the digitalization of requirements has led to the auto-creation of specifications and test plans.

Unique Considerations for SE within Oil and Gas

One of the challenges when considering SE in O&G is that it is difficult to evaluate the entire domain as one. The long and complicated supply chain includes diverse and segmented companies across the globe. And it is not solely crude oil or natural gas. With the current energy transition affecting all aspects of engineering, most O&G companies have been shifting focus from fossil-based systems to include renewable sources that are efficient solutions with net-zero emissions. Many oil and gas companies have targets of net-zero greenhouse gas emissions for operations by 2050.

Another challenge for the domain is that since SE has not been implemented as a holistic approach, there are pockets of SE maturity that do not always intersect. This is a result of most O&G companies following a well-established and practiced sequential (waterfall) stage gate process. Therefore, SE implementation is generally only implemented where a clear case for change is needed and demonstrated, or when all other approaches have been exhausted. To help with this, the INCOSE O&G working group developed a scalable presentation for various high-level conversations and presenting success case studies from participating O&G companies. One area where SE continues to gain traction and show value is in new product development projects. By introducing SE principles and methods early in

the project, the project team can see the benefits of applying SE and embrace the changes to the traditional ways of working.

4.4.7 Power & Energy Systems

Overview of SE Applications within the Power and Energy Domain

During the first two decades of the twenty-first century, the global energy system has been subject to a complex set of requirements stemming from the Paris Agreement, the United Nations Sustainable Development Goals (see [Section 3.1.10](#)), reduction in greenhouse gas emissions, and other efforts to avoid degradation of our social foundations and ecological ceiling (Raworth, 2017). To provide an effective solution for such a complex set of problems demands the realization, or modification, of many new systems, elements, and enabling systems supported by a holistic systems approach. The United Kingdom's (UK) Council for Science and Technology stated with respect to the UK's Net Zero ambitions that “by drawing on SE principles, a detailed and credible plan can provide the framework required to drive change, give reassurance to businesses, investors and consumers, and engage the whole of society in delivering this change” (CST, 2020).

At the heart of the sustainability transition is the convergence of business, technology, and socio-politics to guide innovation around what is viable, feasible, and desirable. This new intersection of disciplines can be enabled through SE. Yet many incumbent organizations and legacy approaches dominate the power and energy landscape, meaning a change in thinking or practice is resisted, or even directly opposed. As a result, the application of SE in power and energy remains largely immature in the first quarter of the twenty-first century compared to more established sectors such as defense, space, and transportation. A cultural shift toward shared knowledge management systems, portfolio management, and organization infrastructure is required to fully embrace and adopt SE.

Unique Terms and Concepts

The language used in SE is made effective through SE heuristics or when translated to real-world examples in the power and energy context. For example, an SoS may be considered an abstract SE term, yet it perfectly describes the nature of distributed energy resources.

Unique Activities, Methods, and Practices

SE may help the realization of effective power and energy systems through the following activities, methods, and practices:

- **Architecture and design.** Provides robust architecture, design, and development processes for higher integrity and interoperability across the supply chain infrastructure from upstream (e.g., solar plant, nuclear, wind farms), mid-stream (e.g., large-scale storage, energy vector processing, transmission and distribution networks) and downstream (e.g., retail outlets, local area networks, domestic microgeneration, private storage).
- **Risks.** Enables better identification and handling of risks associated with energy security and resilience.
- **Portfolio management.** Provides the platform for joined up roadmaps and communication channels which enables stakeholder acceptance, transition, and utilization of emerging paradigms such as smart grids, district heat networks, renewable technologies, demand side response, and electric vehicles.
- **Sustainability.** Enables better management of reductions in greenhouse gas emissions through robust technical and management processes with a whole life cycle perspective, open industry standards, quality management, and assurance.

Other Unique Considerations

Power and energy systems are typically at the SoS level, so activities follow the key characteristics and challenges associated with an SoS. As an

example, achieving the goal of energy security requires as much understanding of geo-politics as it does the evolution of cybersecurity as digitalization grows.

The application of SE needs to transcend geographic boundaries and domain silos. For example, consideration for how SE supports the implementation of the Clydebank declaration, which calls for the establishment of green shipping corridors for zero-emission maritime transport between shipping ports, presents an energy, transport, and logistics challenge on an international scale.

On a global scale, power and energy systems must remain persistently operational for billions of system users whilst maintaining a constant state of equilibrium with demand balanced by supply (augmented by flexibility solutions such as demand side response and storage). In addition, power and energy systems typically have a life cycle of 30–100 years or even into the thousands of years for end-of-life decommissioning and waste storage from nuclear fission facilities. These considerations form complexity multipliers for the sustainable energy transition.

Adapting the mental models and behaviors of system users will be crucial to effecting change. This demands elements of social sciences, systems science, and systems thinking to complement the rigor of SE processes. To support this, there is a need to provide feedback mechanisms to influence the micro-behavior of the human actors in the system in such a way as to maintain the macro-stability of the system (Sillitto, 2010) achieved through HSI (see [Section 3.1.4](#)). But SE cannot focus on change in human behaviors without consideration for market dynamics and political levers. SE can help provide the coherence and joined-up thinking necessary to make energy policy an enabling system for delivering the overarching goals of energy decarbonization, digitalization, decentralization, and democratization. We must also avoid the trap of pushing technology solutions that deliver undesirable user experience. Considerations range from consumer price point, to acoustic noise of technologies, to retrofit disruptions, to the availability of energy in remote or isolated communities. Our challenge, as

future ancestors, will be to find ways to support growing energy demands whilst delivering a sustainable energy supply chain that is available and affordable to everyone on a global scale.

4.4.8 Space Systems

Overview of SE Applications within the Space Domain

Space systems are systems that are designed to operate and perform tasks into and within the space environment. This may consist of: spacecraft (and their associated payloads and instruments); mission packages(s); ground stations; data links between spacecraft and ground, launch systems; and directly related supporting infrastructure. Due to the relatively high costs of deploying assets into earth orbit or beyond, space systems typically require high reliability with little maintenance other than software changes (note that designing for maintainability in space one of the many trade-offs that need to be considered during conceptual design). This makes it necessary for all system elements to work the first time or be compensated by operational workarounds; this can impact the risk posture of the system being developed.

The space domain has evolved into three main areas of interest, with some overlap:

- Civil,
- Commercial, and
- National Security Space.

Each of these areas have their own motivations that can influence the way they develop systems.

Unique Activities, Methods, and Practices

Key emphases of SE in the space domain are integration, verification (including testing), and validation of highly reliable, well-characterized systems. Risk management is also key in determining when to incorporate

new technologies and how to react to changing requirements through multiyear developments and programmatic challenges. SE provides coordination for multi-disciplinary engineering expertise that enables optimized designs.

Civil systems are typically acquired by government agencies, which typically focus on performance risk, determining when to incorporate new technologies, and how to react to changing requirements through multiyear developments and programmatic challenges. This lends itself to the use of the sequential approaches, such as the SE Vee model, or, in some cases, the waterfall model.

Commercial systems strive for profitability (cost and schedule) and are more amenable to using incremental and evolutionary approaches. This allows them to deploy systems faster, and rapidly gain experience that can improve later iterations of their product.

National Security Space, much like Civil, may emphasize performance over cost and schedule, and have traditionally used the Vee and waterfall models for development. They typically are more tolerant of accepting risk from the injection of new technologies.

Unique Standards

Overall, proper application of SE in the space domain helps in design and development of space elements for easier manufacturing and lowering maintenance cost using well-defined processes and standards. SE offers well-defined architecture and design processes and roadmaps for proper interfacing, networking, and communications that enable better integrity and interoperability of space systems and elements provided by various contractors, and across the supply chain using open industry standards.

Civil, commercial, and national security entities have their own drivers that determine how standards are created and adopted. Most space-faring

nations and international consortiums have developed and adopted their own standards that are specific for space systems. Some examples include:

- In the United States, the Department of Defense has created Military (or “Mil”) Standards that have been readily adopted by both Civil and Commercial primarily due to the need for high reliability and survivability in a hostile environment. NASA has also created a set of technical standards, as has the AIAA and other organizations.
- The European Cooperation for Space Standardization is an initiative established to develop a coherent, single set of user-friendly standards for use in all European space activities.
- The Euro-Asian Council for Standardization, Metrology and Certification, a regional standards organization operating under the auspices of the Commonwealth of Independent States, has developed GOST (Russian: OCT), a set of spacecraft certification standards that is commonly used by Russia.
- JAXA (Japan Aerospace Exploration Agency) has developed a library of standards known as JERG (JAXA Engineering Requirement, Guideline).
- International organizations such as ISO and IEEE have also been involved in the development of standards that enhance interoperability.

Other Unique Considerations

An additional challenge in the space domain occurs when humans are integrated into the system. Typically, it includes the incorporation of design features and capabilities that accommodate human interaction with the system to enhance overall safety and mission success. The system needs to ensure that the human needs are addressed in terms of effectively utilizing human capabilities and performance, hazards are controlled to a level considered safe for human operations, and provide, to the maximum extent practical, the capability to safely recover the crew from hazardous situations. At the time of this writing, only missions led by three nations (Russia, the United States, and China) have sent humans into space. Each country has developed their own set of standards:

- Roscosmos for Russia, and
- Human Space Flight Requirements for Civil, typically governed by NASA, for the United States,
- CNSA for China.

Of the three, only the United States has begun to explore human commercial space flight, where such requirements are governed by the US Federal Aviation Administration (FAA).

4.4.9 Telecommunication Systems

Overview of SE Applications within the Telecommunication Domain

Telecommunication systems are defined by having a route to transfer information across and to distinct endpoints that are used to share (send and/or receive) information. They differ from postal systems in that the information shared is in the form of electronic media (applications or services) rather than transporting packages or handwritten letters.

Telecommunication systems are enablers for other services. Almost all modern systems either make use of telecommunication technologies provided by other systems, or contain telecommunication technologies within them (e.g., digital signage and ticketing systems within public transport systems; battlespace communication systems used by the military; environmental monitoring systems such as those used to monitor/predict the weather or detect and provide advance warning of earthquakes and other environmental events). Lives and livelihoods depend on telecommunication systems.

Communication network complexity and the social cost of telecommunication failures will only increase (White and Tantsura, 2016). It is therefore opportune for telecommunication leaders and practitioners to advocate, apply, and extend the best telecommunication SE approaches to cope with this complexity and risk.

Unique Terms and Concepts

Telecommunication systems are built on a wide range of technologies: satellite communication, cellular networks, land mobile radio, microwave, radio, television, Wi-Fi, Bluetooth, and global positioning systems. They are increasingly software-intensive systems (Donovan and Prabhu, 2017).

Telecommunications includes communication networks owned by carriers, internet service providers, government agencies, and other enterprises, as well as broadcast networks (e.g., radio, cable, television) and over-the-top service provider applications (e.g., messaging, video conferencing, social media applications) (Adkins, et al., 2020) (Birman, 2012). Some telecommunication systems, like the internet and the public switched telephone network, have no single owner; their design depends upon collaboration in international telecommunication standards bodies.

The telecommunication transport network may be dedicated for a specific purpose (service or application) or shared for multiple services or applications. Communication networks may be employed for emergency services, defense, transportation, health, financial, industrial supervisory control, and data acquisition purposes. Many of these are considered to be national critical infrastructure (Lewis, 2019).

Telecommunication systems typically have some of the following characteristics:

- Diverse geographical distribution;
- Multi-party ownership and management (network domains or applications);
- Multiple constituent systems with independent life cycles that continuously evolve over many decades;
- A small stable set of functions, allocated across system elements, to achieve the common purpose of enabling communication;
- Many nodes and types of nodes; and
- Strong interdependence between nodes (failures within one node may cause other nodes to become isolated unless the specific failure mode is

anticipated and the network is designed to withstand the failure).

Unique Activities, Methods, and Practices

Network specifications, architectures, and models have a small number of functions and node types at their core, but they must also support many function and node variants. Engineering planning and architecture must be flexible enough to accommodate change in parts of the network owned and operated by others. Design and verification activities depend on a thorough analysis of failure modes and interactions across the network. Scale and variation lead to significant configuration management challenges (Xu and Zhou, 2015). These characteristics affect engineering activities throughout the life cycle. Like other types of networks, communication networks can be represented by a network model that defines a grouping of nodes and links to help understand how resources flow from one node to another.

While some telecommunication systems in slowly changing environments can survive on implicit engineering practices, such approaches have been found to be ineffective and inefficient. They typically fall short when one or more of the following exist:

- New and/or complex stakeholder needs;
- New operating models;
- Significant safety or security risks;
- High complexity; or
- Constrained, high-cost operating environments.

Other Unique Considerations

Many vendor services and technologies are mature and are slow to evolve. Telecommunication networks typically comprise COTS vendor equipment/applications using industry interface standards and semi-standardized architectures (see [Section 4.3.3](#)). This equipment is typically integrated together without the use of SE. The promise from COTS vendors is that their equipment is suitable for rapid configuration and integration,

and implicitly can survive with simpler requirements/business analysis and engineering processes, including outsourced vendor standardized engineering. This can lead to situations where there are unexpected outcomes. Without an appropriate set of engineering disciplines, such as those based on SE, it is difficult to assess, let alone manage, the risks.

4.4.10 Transportation Systems

Overview of SE Applications within the Transportation Domain

Ground transportation systems, such as highways, busses, people-movers, mass transit, and rail involve complex capital programs for fleet acquisition and/or building of related infrastructures and are invariably within a SoS on an operational level. The system life cycle for ground transportation assets is 25 to 100+ years and often involves public funding and a related fiducial public trust. During this life cycle, operational processes are continuously optimized and improved using block changes.

The ground transportation industry segment is an emerging SE practice area, largely consisting of transit authorities, railroad operators rolling stock manufacturing industries, and civil engineering construction firms. Globally, some geographic regions are progressing the deployment and acceptance of SE more rapidly. They have created effective SoS approaches that are in concert with emergent societal trends such as smart cities, an embedded safety culture, and a through life approach to service delivery. For example, in the UK, the Institution of Civil Engineers has made solid progress in moving toward an SE approach within the civil engineering domain. Other regions are at the early stages of integrating SE into their ground transportation systems processes. These imbalances can be overcome in time through global collaboration.

Unique Considerations

Additional mandates for the use of SE methods, skills, and competencies are generated from rapidly increasing system complexities in modern

transportation systems, as evidenced by Intelligent Transportation System (ITS) concepts including regional, national, and local smart cities initiatives. Train control automation, bus scheduling, and ride-share coordination, coupled with autonomous vehicles, micro-grid hybrid traction power, passenger fare collection, and related trip planner smartphone apps provide emerging complexities within modern ground transportation capability. However, public transportation must improve as public needs change, new technologies are introduced, or as environmental concerns and quality-of-life concerns encourage greater use of public transit. Ground transportation services are typically “in-service” brownfield systems and must migrate to an updated version, while still maintaining service, so that assets may be cost effectively managed in the public’s interest (see [Section 4.3.2](#)). Some transit assets in large cities run 24/7 service as a matter of public safety and demand, requiring careful planning. A “whole” organization approach is needed for success.

Unique Terms and Concepts

The transportation domain uses the same terms as the infrastructure domain, see [Table 4.7](#).

Unique Activities, Methods, and Practices

Some examples of early initiatives on the application of SE are in the United Kingdom, the European Union, and, specifically, the Netherlands. In the UK, Network Rail established the Governance for Railway Investment Projects (GRIP) to help manage projects and to align them with SE processes. Leading EU train operating companies and the main rolling stock manufacturers seek to develop a common, open architecture such as EULYNX and the Reference Command and Control System Architecture to support and optimize trackside and rolling stock acquisitions and upgrades. In the Netherlands, SE is advocated by the network management institution, ProRail, as part of their project management approach for delivering rail infrastructure projects. ProRail created a SE Handbook for their project life cycle approach which is comparable to GRIP.

5

SYSTEMS ENGINEERING IN PRACTICE

5.1 SYSTEMS ENGINEERING COMPETENCIES

The terms “competence” and “competency” are two distinct terms used to define the personal attributes of individual Systems Engineering (SE) practitioners. Competence is the ability to perform an activity or task. Competency is the set of skills required in the performance of a job. The competence reflects the total capacity of the individual, whereas a competency is a set of skills that the individual will be required to perform for a job. The sum of an individual’s competencies will make up their competence. Those competencies are measured and assessed to provide an estimate or a picture of the overall competence of a SE practitioner.

The INCOSE Systems Engineering Competency Framework (SECF) (2018) provides a set of competencies that identify knowledge, skills, abilities, and behaviors important to effective SE that can be applied in any domain context. The INCOSE SECF spans a wide range of competencies organized into five themes: core, professional, technical, management, and integrating.

- The *core competencies* underpin engineering, as well as SE.
- The *professional competencies* are primarily based in behavior as a SE practitioner (see [Section 5.1.2](#)).

- The *management competencies* relate to performing tasks associated with controlling and managing SE activities.
- The *technical competencies* are associated with technical processes to accomplish SE.
- The *integrating competencies* recognize SE as an integrating discipline, considering activities from other specialists from project management, logistics, quality, and finance to create a coherent whole.

The INCOSE Systems Engineering Competency Assessment Guide (SECAG) (2022) provides guidance on how to evaluate individuals for proficiency in the competencies and how to differentiate between proficiency at each of five levels defined within the INCOSE SECF. For each competency, the INCOSE SECAG provides a description, why it matters, and possible contributory evidence. Indicators of competence are provided along with examples of relevant knowledge, experience, and possible objective evidence of personal involvement in activities or professional behaviors applied.

Both the INCOSE SECF and INCOSE SECAG are comprehensive resources that are globally accepted and tailorable to the needs of the organization or individual. The INCOSE SECF and INCOSE SECAG are intended for use in hiring, assessing, training, and advancing SE practitioners; and may be used for other purposes as deemed appropriate.

Given the complexity and multidisciplinary nature of today's systems and system of systems, it is not possible for a single person to know everything about a system of interest (SoI). As shown in [Figure 5.1](#) SE practitioners are often referred to as “T-shaped” in describing their professional expertise (Delicado, et al., 2018). SE practitioners must have both a depth of knowledge of a fundamental engineering discipline, while at the same time develop and maintain a breadth of knowledge about systems and the multiple disciplines involved.

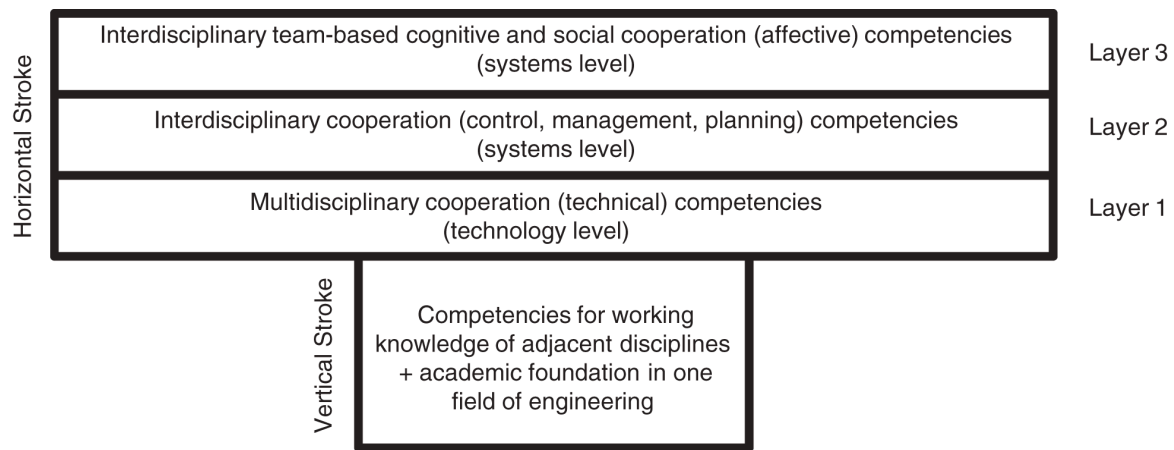


FIGURE 5.1 The “T-shaped” SE practitioner. From Delicado, et al. (2018). Used with permission. All other rights reserved.

5.1.1 Difference between Hard and Soft Skills

“The complexity of modern system designs, the severity of their constraints, and the need to succeed in a high tempo, high-stakes environment where competitive advantage matters, demands the highest levels of technical excellence and integrity throughout the life cycle” (INCOSE SECF, 2018, Page 47). Interactions with stakeholders, including customers, project managers, all types of engineers, operations, marketing, and various departments contribute to the overall success of a system and to the life cycle cost (LCC) to conceive, produce, utilize, support, and retire that system. SE practitioners and SE teams need both hard skills and soft skills to meet these challenges. [Table 5.1](#) contrasts the differences between hard skills and soft skills. As examples in the hard skills category, consider typical engineering technical aspects like structural, hydrodynamic, reliability, or electrical analyses, because they require science, mathematics, and quantitative modeling in order to solve engineering-related problems. Examples of hard skills for SE practitioners include requirements analysis, architectural evaluation, and risk management. As examples in the soft skills category, consider influencing a peer, motivating a team, and resolving a conflict. They require emotional intelligence and appropriate behaviors to solve people-related problems.

TABLE 5.1 Differences between the hard skills and soft skills

Hard Skills	Soft Skills
• Concrete definitions • Measurable • Testable • Individual application • Low self-awareness required • Personal affect has low impact	• Subjective definitions • Difficult to measure • Difficult to test • Social application • High self-awareness required • Personal affect has high impact

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Soft skills, which are also known as interpersonal skills, intrapersonal skills, people skills, professional skills, and other terms, are those skills related to aspects such as teamwork, collaboration, and facilitation. Understanding the nuances of an organization, the dynamics of a team, or the experiences of an individual requires soft skills. It is not uncommon for an engineer to question the value of soft skills, and question how these skills apply to a specific engineering problem. However, soft skills often make the difference between a smooth application of SE and one riddled with challenges. Both the INCOSE SECF and INCOSE SECAG outline areas of professionalism and ethics that provide insights and information related to important soft skills for ensuring long-term sustainability for the life cycle of a system.

The field of engineering brings a unique perspective toward developing competencies in these hard and soft skills. Early in the career of an engineer, there is a strong emphasis toward competency in the hard skills,

such as those related to engineering, math, physics, chemistry, industrial processes, and technical management. The INCOSE SECF outlines these skills in the areas of Technical Competencies, Management Competencies, and Core Competencies. As the career of an engineer develops, the emphasis moves toward a balance between hard and soft skills. For an SE practitioner, a higher demand is often placed on these soft skills due to interactions with stakeholders, team members, and senior managers. The INCOSE SECF outlines these skills in the Professional Competencies.

5.1.2 System Engineering Professional Competencies

Professionalism can be summarized as a personal commitment to professional standards of behavior, ethics, obligations to society, the profession, and the environment. SE practitioners are trusted to apply reasoning, judgment, and problem solving to reach unbiased, informed, and potentially significant decisions because of their specialized knowledge, skills, abilities, and behaviors. SE professionalism includes consideration of personal behaviors beyond using methods and tools. SE practitioners recognize the benefits of behaviors and outcomes related to professional competencies from ethics, professionalism, and technical leadership to communications, negotiation, team dynamics, facilitation, emotional intelligence, coaching, and mentoring. These Professional Competencies are documented in the INCOSE SECF (2018). The evaluation of an individual's Professional Competencies can be accomplished using the INCOSE SECAG (2023).

5.1.3 Technical Leadership

Leadership can be generally defined as: “The act (or art!) of enabling people to produce results or achieve outcomes they would not have on their own.” Technical Leadership is leadership in situations that involve technology. As illustrated in [Figure 5.2](#), Technical Leadership exists at the intersection of Technical Expertise and Leadership Skills.

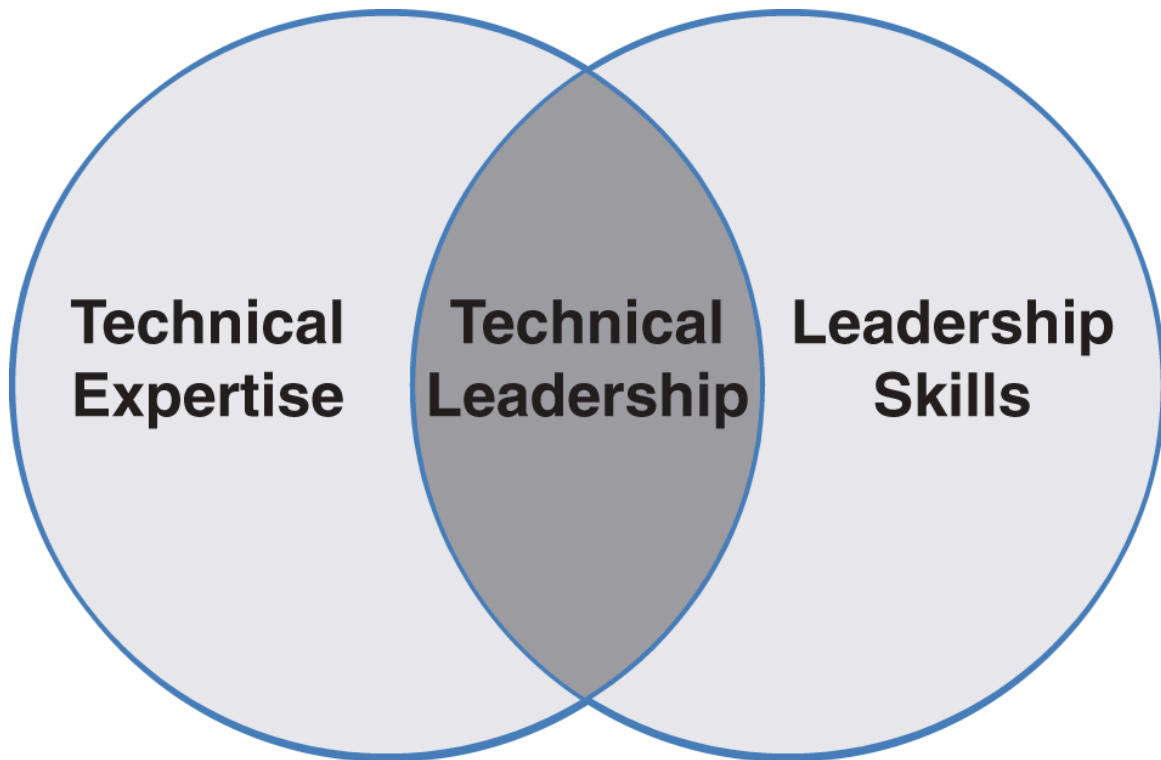


FIGURE 5.2 Technical leadership is the intersection of technical expertise and leadership skills. INCOSE SEH original figure created by Gelosh and Pennotti on behalf of the INCOSE Technical Leadership Institute. Usage per the INCOSE Notices page. All other rights reserved.

Strong technical leadership is critical for the successful development, operation, sustainment, and evolution of engineered systems. To successfully lead technical teams and technical enterprises, technical leaders must possess all the leadership skills required of any effective leader. Technical leaders must also possess, and be recognized as possessing, significant technical expertise. A good technical leader must possess expertise in one or more technical areas and have some level of understanding across a wide range of disciplines to earn credibility for leading technical teams. They must also be aware of the limits of their own knowledge so they know when to seek the expertise of others.

Why Is Technical Leadership Important for SE Practitioners?

SE practitioners are responsible for the success of the system as a whole. They must understand the needs of a broad range of stakeholders and ensure those needs are met. They must work across traditional engineering disciplines to ensure the individual contributions of each integrate harmoniously to produce the desired outcome. Because they seldom have the positional authority to ensure these outcomes, they must lead through

influence, leveraging their technical knowledge and their personal qualities to create an environment in which the individuals and teams accomplish the desired goals.

What Does It Take to Be a Good Technical Leader?

The INCOSE Technical Leadership Institute (TLI) developed a model that identifies and describes six interrelated behaviors that technical leaders must master to successfully lead through influence. Each behavior is described in [Table 5.2](#), and each includes a question technical leaders should continually ask themselves as they seek to lead.

TABLE 5.2 Technical leadership model

Technical Leadership Behavior	Description	Question for the Technical Leader
Holding the Vision	A vision is an aspirational statement that defines who we are and where we want to go. It provides an impelling purpose. It must be the start of a continual and ongoing conversation that SE practitioners are well positioned to support, reinforce, and encourage.	<i>What outcome are we striving to achieve and how am I advancing that vision?</i>
Thinking Strategically	Using Strategic Thinking, technical leaders formulate a hypothesis before acting, treat the action as an experiment to test their hypothesis, and, based on the results they observe, continue along the path they are on or formulate a new hypothesis and begin the testing anew.	<i>What patterns are emerging from my actions, and what are the implications for what I should do next?</i>
Fostering Collaboration	Complex problems cannot be solved by individuals working alone; their solution requires the efforts of many. The goal is to foster collaboration that allows new ideas to emerge	<i>What relationships am I building today for myself and for others?</i>

Technical Leadership Behavior	Description	Question for the Technical Leader
	through creative conflict and experimentation.	
Communicating Effectively	Effective communication is not just about speaking, it is also about actively listening which requires attention to both the content being spoken and the emotion with which it is communicated.	<i>Who am I trying to influence and what is their greatest challenge?</i>
Enabling Others to Succeed	The technical leader's role is to influence, guide, encourage, and support those who can produce desired outcomes. The leader's success will derive from their success, and their acknowledgment that the leader contributed to it.	<i>What obstacles are preventing others from acting, and how can I help remove those obstacles?</i>

Technical Leadership Behavior	Description	Question for the Technical Leader
Demonstrating Emotional Intelligence	In order to lead others, technical leaders must continually seek feedback to decrease their blind spots and be willing to reveal things that help others know them better. While the former can make them uncomfortable and the latter make them more vulnerable, the payoff will be more than worth the effort.	<i>That am I afraid of admitting to others and how might disclosing it improve our relationship?</i>

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5.1.4 Ethics

There will always be pressure to cut corners to deliver projects faster or at lower costs, especially for a profession such as SE. As stated in the INCOSE Code of Ethics (2023),

The practice of SE can result in significant social and environmental benefits, but only if unintended and undesired effects are considered and mitigated.

Part of the role of the SE practitioner as a leader and professional is knowing when unacceptable risks or trade-offs are being made, knowing how to influence key stakeholders, and having the courage to stand up for stakeholders, the community, and the profession when necessary. The

INCOSE Code of Ethics contains sections on “Fundamental Principles,” “Fundamental Duties to Society and Public Infrastructure,” and “Rules of Practice” to help the SE practitioner in practical applications of ethics to their work and daily lives.

5.2 DIVERSITY, EQUITY, AND INCLUSION

The following definitions are taken from the Accreditation Board for Engineering & Technology (ABET) (2017) and provide a reference point for conversations and materials about diversity, equity, and inclusion.

- *Diversity* is the range of human differences, encompassing the characteristics that make one individual or group different from another.
- *Equity* is the fair treatment, access, opportunity and advancement for all people, achieved by intentional focus on their disparate needs, conditions, and abilities.
- *Inclusion* is the intentional, proactive, and continuing efforts and practices in which all members respect, support, and value others.
- INCOSE uses the compound term Diversity, Equity, and Inclusion (abbreviated to DEI) when referring to the broad subject matter.

Diversity encompasses a wide range of characteristics. [Figure 5.3](#) shows a representative set of characteristics grouped into five areas: intrinsic, employment, environment, interaction, and family (Harding and Pickard, 2019).

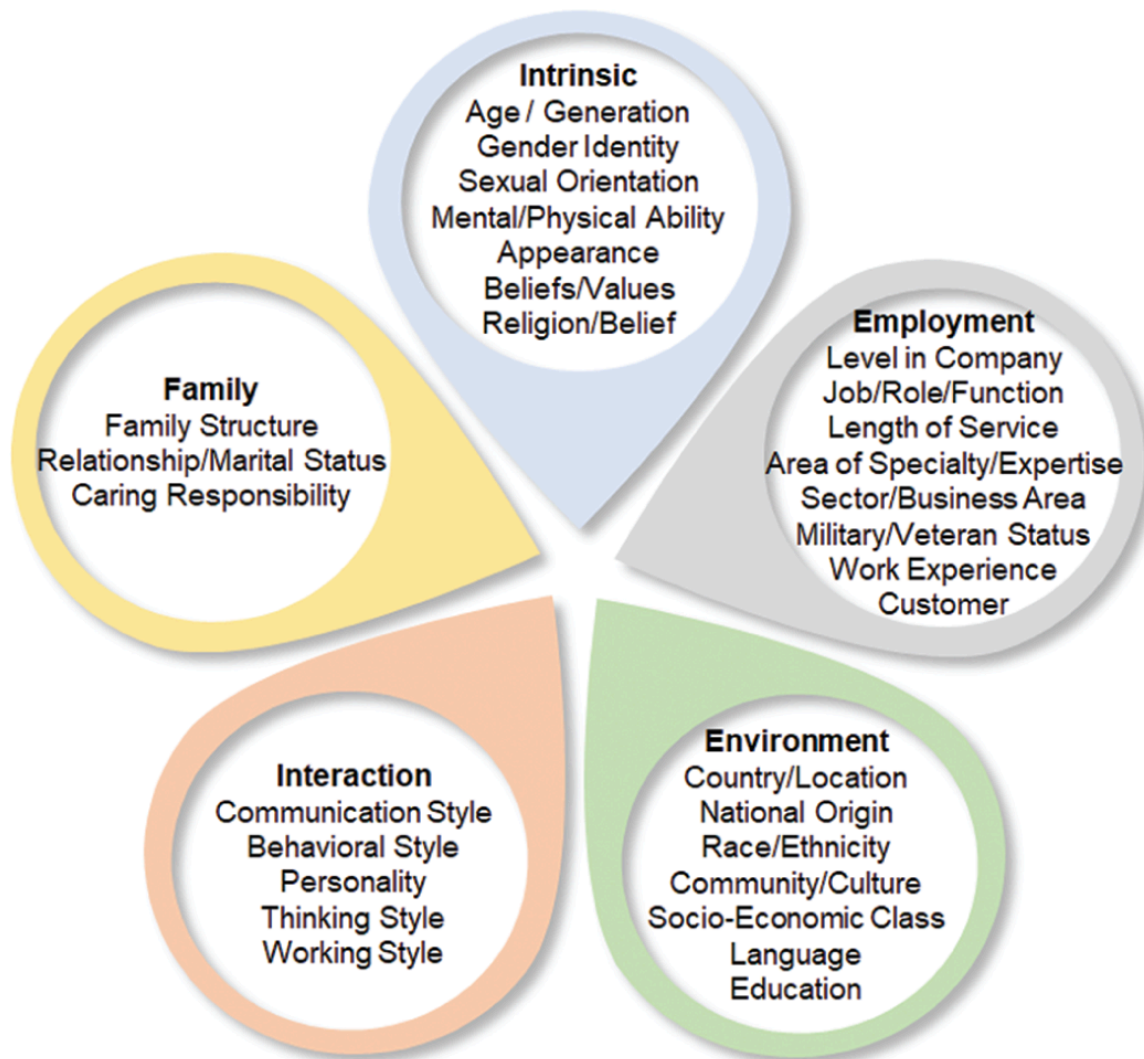


FIGURE 5.3 Categorized dimensions of diversity. From Harding and Pickard (2019) derived from SEBoK (2023). Used with permission. All other rights reserved.

DEI is vital to successful SE because of the wide range of contexts in which SE is applied and the consideration of multiple stakeholder viewpoints at the heart of the systems approach. SE practitioners play a pivotal role in integrating DEI concepts into the team's composition and approach and in the system design and development process through:

1. Ensuring that the SE team and its leadership is inclusive, welcomes a diverse range of talent, promotes cognitive diversity and diversity of ideas, and, where necessary, takes deliberate action to provide psychological safety and communication equity.
2. Ensuring that the systems we realize are as accommodating as possible of the differences within the entire stakeholder community.

Failure to address either of these aspects results in sub-optimal outcomes, whether in terms of missed solutions, lower productivity, or delivering a system that does not equitably meet the needs of the full range of the stakeholder community (i.e., essentially failing to meet the ultimate goal of delivering a total optimal system solution for all). Furthermore, in providing total optimal system solutions, one purpose of the system is to provide similar (equal) outcomes to each user. To accomplish this, the SE practitioner must understand the difference between equality and equity. Equality is providing each user exactly the same resources but may not result in similar (equal) outcomes. Equity, however, is proactively addressing disparities that exist between individual users (e.g., different situations, needs, requirements, life experiences, challenges) and changing the system so that each user can experience a similar outcome.

5.3 SYSTEMS ENGINEERING RELATIONSHIPS TO OTHER DISCIPLINES

SE practitioners routinely work within broad multidisciplinary teams. The following sections highlight SE practitioner interactions with some key related disciplines.

5.3.1 SE and Software Engineering (SWE)

This section describes an overview of the relationships between SE and software engineering (SWE), especially when developing and modifying *software-intensive systems*. ISO/IEC/IEEE 42010 (2022) defines a software-intensive system as:

Any system where software contributes essential influences to the design, construction, deployment, and evolution of the system as a whole.

Section 4.3.4 discusses software-intensive systems. These systems are also known as software-enabled and cyber-physical systems (see [Section 4.3.5](#)).

What is this discipline?

ISO/IEC TR 19759 (2015) defines SWE as:

The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software.

SE practitioners need to know that software has become ubiquitous in modern systems and is important in updates to many older systems. Because software elements are malleable, software can thus, in most cases, be tailored, adapted, and modified more readily than can physical elements, especially in the support stage of the system life cycle. Software elements, like physical elements, can be improved and replaced if the interfaces are preserved. Software often provides or facilitates provision of the following system attributes: functionality, behavior, quality characteristics (QCs), and system interfaces (both internal and external). Software embedded in software-intensive systems can also monitor system performance and provide observable performance indicators.

The differences between software elements and physical elements present significant challenges for SE practitioners when facilitating development and modification of software-intensive systems. Software elements are logical entities composed of textual and iconic symbols that are processed and interpreted by other software that is executed on computer hardware. A single mistyped symbol, including a single mistake in a logical expression, in large system that may include thousands or millions of symbols and logical expressions can result in failure of the entire system. The mistake may not be detected during system development or modification because even small software programs contain large numbers of logical branches and iteration loops that prevent exhaustive testing in reasonable amounts of time. Assuring the quality of a software-intensive system is thus challenging. Detailed concentration on the precise development of software elements and software interfaces may cause SWE

practitioners to lose sight of the impacts of their software on larger system issues.

What is its relationship to SE?

The relationship of SWE to SE is becoming increasingly important because software is a large and growing part of modern systems. The issues that arise for *SE* practitioners when developing or modifying software-intensive systems involve partitioning of requirements and architecture so that SWE practitioners (and hardware engineering (HWE) practitioners of various kinds) can design their elements and interfaces and pursue their development practices based on the differing natures of the mediums in which they work (Fairley, 2019). Well-defined and controlled interfaces, both provided and required, are essential so that separately developed elements can be efficiently integrated. Integration of software and hardware elements is typically accomplished in an incremental manner, as the elements become available. Defining and controlling interfaces is an ongoing challenge.

The primary benefit of coordinating SE and SWE during system development and modification is incorporation of software-provided capabilities in a manner that results in efficient and effective systems. SE practitioners can consult with SWE practitioners during system analysis and design to develop options and tradeoffs for configuring software and hardware elements and their interfaces. In addition, SWE practitioners can provide recommendations to SE practitioners for processes that can be used when developing and integrating software and hardware elements and when performing system verification, validation, and deployment (Fairley, 2019). Consultations and recommendations can ameliorate the software problems that sometimes result in late deliveries, insufficient system performance, and difficult system modifications.

How does it impact/is it impacted by SE?

SE practitioners who develop software-intensive systems are, as always, concerned with facilitating development of systems that are delivered in a timely manner, that satisfy performance parameters, and that can be modified efficiently and effectively. SE practitioners can better achieve these goals by taking advantage of the expertise of SWE practitioners, but they may fail to do so because they may not be familiar with the culture, terminology, and practices of SWE. SWE practitioners may not contribute their expertise because they are not consulted or if consulted may have insufficient knowledge of system level issues to provide recommendations. Involving knowledgeable SWE practitioners at the system level can improve communication. Cross-training, shadowing, mentoring, and collaborative workshops that include SE practitioners and SWE practitioners can result in synergetic relationships that will provide better communication and allow more effective and efficient development and modification of software-intensive systems.

More information on SE for software-intensive systems can be found in The Guide to the System Engineering Body of Knowledge (SEBoK) in the Part 6 knowledge area titled “SE and Software Engineering.”

5.3.2 SE and Hardware Engineering (HWE)

This section describes an overview of the relationships between SE and hardware engineering (HWE).

What is this discipline?

HWE includes the development and implementation of physical elements for systems, enabling systems, and support equipment for systems. HWE includes mechanical engineering (ME) for mechanical elements and electrical engineering (EE) for electrical and electronic elements.

What is its relationship to SE?

The SE team must assist the hardware team in establishing hardware requirements, physical interface requirements, and establishing and

tracking key physical measures (e.g., size, weight, and power (SWaP) budgets) at every level of the system architecture. During the System Architecture Definition and Design Definition processes (see [Section 2.3.5.4](#) and [2.3.5.5](#)), requirements will be allocated and derived across many hardware elements. Initially, these allocations may be required to meet a set of requirements at the system level without a full understanding of the actual values at the element level. This is where a requirements budget can be utilized by the SE team to derive these critical hardware attributes across all the applicable hardware elements and design teams. These hardware design teams will consist of diverse HWE fields, with a focus on the hardware design. The hardware design teams will accept an initial budget allocation from the SE team based on a summary analysis of the system requirements and design. The SE team gives allocations based on the requirements and hardware design team inputs. The SE team will then track progress to those budgets, determine any impacts at the system level, and relay change requests as needed. Hardware considerations include SWaP, inertia, balance, frequency, phase, and others.

Interface management is an important consideration for SE and HWE. Interface decisions will have an impact on both the interconnectivity of the system and hardware selection. Utilizing open source or standard interfaces, if feasible, can reduce costs and development time. If unique or custom interfaces are necessary, they must be documented in the architecture before design, development, and testing can occur. This requires the SE team to balance interface decisions between capability and hardware standards.

The application of SE to hardware differs from software-only applications in two primary ways: hardware solutions may exist that can meet all or part of a decomposed requirement and the hardware performance requirements generally must be built into the initial hardware deliverable rather than iterated into the design (with some exceptions). A make or buy decision is often performed at each level of the architecture based on the results of a market analysis and tradeoff study of existing solutions that will satisfy the requirements. If the result of the decision is to make, the

architecture can be further decomposed or Implementation process initiated. However, if the result of the decision is to buy, then the architecture decomposition for that system element will end (see [Section 1.3.5](#)).

How does it impact/is it impacted by SE?

Hardware material selection is supported by existing application or domain-specific standards. On a smaller, less complicated systems, HWE may be able to select all materials without the need of a formal SE process. With the increasing complexity of large systems and SoSs, the role of SE in hardware material selection becomes paramount, especially in applications where human lives could be affected. For instance, when multiple hardware sources could have system elements with mechanical or electrical interactions, the role of SE is to document architecture and design decisions and provide timely input into the material selection process before costly errors are found in verification or production. Sometimes SE acts to balance the software and hardware requirements, because software requirements can influence the hardware requirements, and vice versa. For instance, having a high availability requirement can result in the need for redundant software systems deployed on redundant hardware systems, meaning that the weight and the power budgets are increasing. SE serves as the bridge between software and hardware.

5.3.3 SE and Project Management (PM)

This section describes an overview of the relationships between SE and project management (PM).

What is this discipline?

As defined by the Project Management Institute (PMI), PM is defined as:

The application of knowledge, skills, tools, and techniques to project activities to meet the project requirements.(PMI, 2022)

PM activities include initiating, planning, executing, monitoring, and closing *projects*. Within this handbook these are primarily distributed across the Technical Management Processes (see [Section 2.3.4](#)) for the SE portion of the responsibilities, but also include some activities in the Agreement Processes (see [Section 2.3.2](#)) and Organizational Project Enabling Processes (see [Section 2.3.3](#)). In the PMI Project Management Body of Knowledge (PMBOK) (2017), they are part of Project Integration Management, Project Scope Management, Project Schedule Management, Project Cost Management, Project Quality Management, Project Resource Management, Project Communications Management, Project Risk Management, Project Procurement Management, and Project Stakeholder Management. According to the PMBoK, a *program* is a set of related projects that are viewed as an entity that requires coordinated management.

What is its relationship to SE?

While SE and PM are distinct disciplines, extensive research has shown that effective integration between PM and SE improves project performance, achieving better results in schedule and budget performance, as well as stakeholder requirements satisfaction, when compared with projects with lesser integration. In 2011, INCOSE formed a strategic alliance with PMI and Massachusetts Institute of Technology (MIT) to research and advance the integration of the two disciplines, driven by the vision that better integration would lead to the delivery of better solutions for organizations and their stakeholders. One output of this alliance was *Integrating Program Management and SE* (Rebentisch, 2017), which defines this integration as “a reflection of the organization ability to combine project management and SE practices, tools and techniques, experience and knowledge in a collaborative and systematic approach in the face of challenges in order to be more effective in achieving common goals/objectives in complex project environments.” A summary of the respective and shared responsibilities between PM and SE is shown in [Figure 5.4](#). Cooperation between project managers and SE practitioners must exist within all these shared activities.

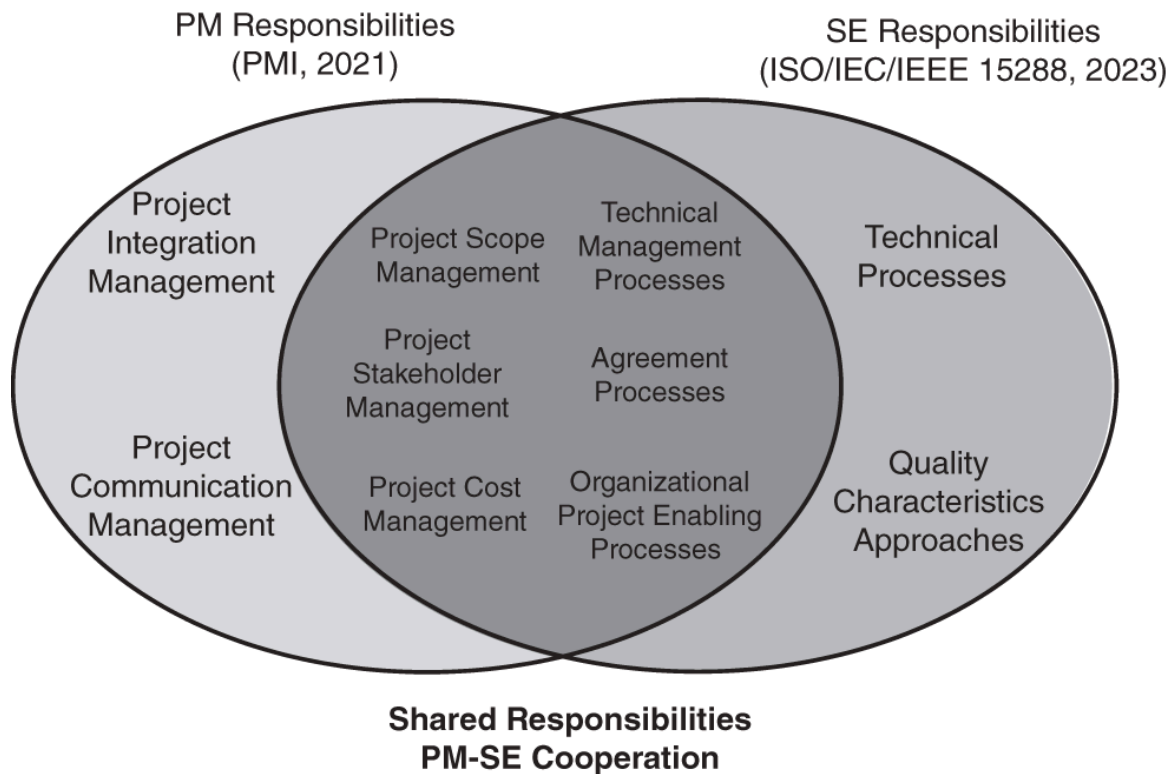


FIGURE 5.4 The intersection between Pm and SE. INCOSE SEH original figure created by roussel on behalf of the INCOSE Pm-SE Integration Working Group. Usage per the INCOSE Notices page. All other rights reserved.

How does it impact/is it impacted by SE?

A high degree of integration between PM and SE is characterized by collaborative decision making, a shared responsibility toward a common goal, having the project manager involved in technical aspects of the project, and having the SE practitioners involved in programmatic aspects of the project. An understanding of the differences, culture, background, and behavior of the two disciplines is also required. A team's ability to combine PM and SE practices, tools and techniques, experience, and knowledge in a collaborative and systematic approach enables addressing challenges in order to achieve common goals and objectives. Specifically, an integrated team achieves rapid and effective decision making, effective collaborative work, and effective information sharing (Rebentisch, 2017).

PM and SE overlap in the early stages of concept and development but tend to diverge in the later stages of development and production. For example, in the early life cycle stages the SE practitioner focuses on the technical details of the SoI, verification, and validation. The project manager focuses

on the overall project performance and delivery of benefits, including high-level finance and budgetary requirements. PM and SE should cooperate on concurrent development of the breakdown structures (see [Section 2.3.4.1](#)), and in the management of them through the life cycle. If the different structures are managed separately by the respective teams without coordination, problems may arise in the project. As the project proceeds through the later life cycle stages, greater integration between PM and SE reduces unproductive tension, a cause of project delays, cost increases, and, sometimes, project failure (Rebentisch, 2017).

As stated in Rebentisch (2017), PM delivers the sustainable benefits of the overall project, while SE delivers the technical aspect of the project. These two roles overlap to integrate technical and programmatic aspects of the project and create potential for unproductive tension, if not effectively managed with cooperation. For example, PM and SE share the same objective to satisfy stakeholder needs and requirements. However, project managers tend to focus on project stakeholders, while SE practitioners tend to focus on system stakeholders. The separation between these two types of stakeholders can generate tension and misunderstandings between project managers and SE practitioners. It is important that they jointly identify all stakeholders and agree on the priority and criticality of stakeholder needs and requirements.

Where there is not effective integration between PM and SE, unproductive tension emerges between the two disciplines. This tension produces conflict and works at cross purposes with project success. Tension can be related to practices, techniques, as well as responsibilities. Misaligned measures can cause tension. Common measures are critical to ensuring that each party has the same concerns and information. The maturity level of each party is critical. An immature or inexperienced PM organization can render ineffective a mature, high-performing SE organization (and vice versa). When one or both disciplines perform inadequately, the entire effort is impaired.

5.3.4 SE and Industrial Engineering (IE)

This section describes an overview of the relationships between SE and industrial engineering (IE).

What is this discipline?

Bidanda (2022) defines Industrial Engineering as:

Optimizing the utilization of human resources, facilities, equipment, tools, technologies, information, and handling of materials to produce quality products and services safely and cost-effectively considering the needs of customers and employers.

The Institute of Industrial and Systems Engineers (IISE) Industrial and SE Body of Knowledge (ISEBoK) (2022) is composed of 14 knowledge areas. The knowledge areas include: Operations Research & Analysis, Economic Analysis, Facilities Engineering & Energy Management, Quality and Reliability Engineering, Ergonomics & Human Factors, Operations Engineering & Management, Supply Chain Management, Safety, Information Engineering, Design and Manufacturing Engineering, Product Design & Development, Systems Design & Engineering.

What is its relationship to SE?

IE is closely related to SE. The IISE states that “Industrial and SE is concerned with the design, improvement, and installation of integrated systems of people, materials, information, equipment and energy. It draws upon specialized knowledge and skill in the mathematical, physical, and social sciences together with the principles and methods of engineering analysis and design, to specify, predict, and evaluate the results to be obtained from such systems.” As evidence of the close relationship of IE and SE, [Figure 5.5](#) compares the ISEBoK (2022) with SE topics in this handbook. The numbered topics on the figure are the knowledge areas explicitly identified in the ISEBoK. The figure identifies the knowledge

areas that are usually performed by SE practitioners, the ones usually performed by industrial engineers (IE), and the knowledge area descriptions that are used by both disciplines.

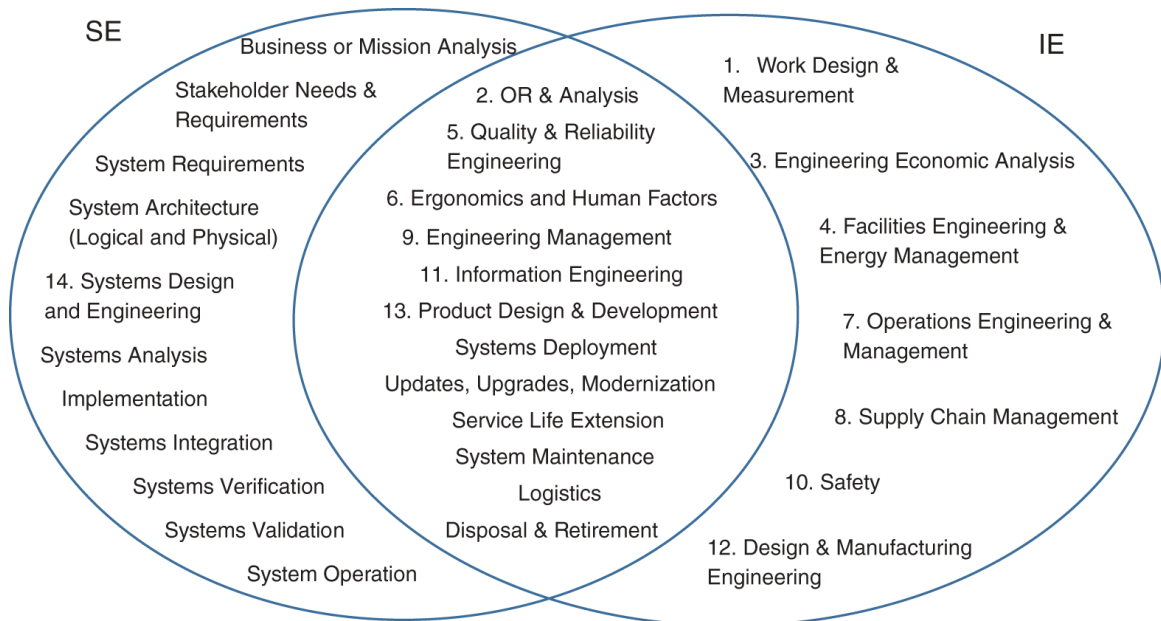


FIGURE 5.5 IE and SE relationships. From Parnell, et al. (2022). Used with permission. All other rights reserved.

As an illustration of the relationships between IE and SE, four knowledge areas are discussed below.

Operations Research and Analysis—Operations Research (OR) includes a variety of techniques to quantify and improve the efficiency of systems and organizational processes using scientific mathematical models. The mathematical techniques include linear programming, transportation models, linear assignment models, network flows, dynamic programming, integer programming, nonlinear programming, metaheuristics, decision analysis, game theory, stochastic modeling, queuing systems, simulation, systems dynamics, and analytics. Industrial engineers use OR to understand, design, and improve the operation of industrial systems and processes. SE practitioners can perform OR analyses or use OR studies performed by others to make system decisions (see [Section 5.3.5](#)).

Information engineering—Information engineering is a “methodology for developing an integrated information system based on the sharing of

common data, with emphasis on decision support needs as well as transaction-processing (TP) requirements” (Gartner, 2022). Information engineering topics include: data types; information system concepts; information requirements; output design; data processing; database concepts; storage and processing; system analysis; system design; system evaluation; information management; and data analytics. Since information systems are a critical component of modern engineering systems, IE and SE practitioners work closely with information engineers.

Product Design and Development—As stated in ISEBoK (2022), “From an Industrial Engineering knowledge view, it is the processes and analysis employed supporting efficient decision-making during Product Design and Development.” Product design and development topics include: design process; preliminary and detailed design; verification and testing; planning for manufacture; metrics for design and development; life cycle costing; and risk and opportunity management. SE practitioners may provide inputs to the product design and development process throughout the system life cycle.

System Design and Engineering—The ISEBoK (2022) lists the SEBoK (2022) as the reference for this section. The topics include the system life cycle concepts from mission engineering to operations.

How does it impact/is it impacted by SE?

There are some significant overlaps. IE and SE practitioners must work closely with each other to accomplish the goals of the project. Different organizations use different job titles for IE and SE, and the same job title may have different responsibilities in different organizations.

5.3.5 SE and Operations Research (OR)

This section describes an overview of the relationships between SE and operations research (OR).

What is this discipline?

The Institute for Operations Research and the Management Sciences (INFORMS) defines OR as:

The scientific process of transforming data into insights using advanced analytical methods to making better decisions.(INFORMS, 2022)

The name “operations research” originated from the recognition of the successful use of scientific and mathematical modeling techniques for military *operations* during World War II (Gass and Harris, 2001). For example, mathematical modeling assisted in developing strategies to employ bombers as well as arrange convoys against submarine attacks. OR is sometimes referred to as Management Science, particularly in business applications, as well as Operational Research in some countries (Quade and Boucher, 1968).

A branch of applied mathematics, OR includes wide range of approaches. Common to all is the establishment of a model to represent the system to support analysis for decision making. In some cases, the model may be solved directly, but other common approaches include optimization, simulation, and decision analysis.

Optimization is useful in determining the best combination of decision variables. Optimization models can be linear, integer, and nonlinear depending on the form of the objective function and constraints. Goal programming is useful when thresholds are established as an achievement target on multiple objectives. The approach avoids falling short of goals and incurring a penalty but permits a broader trade-off analysis.

Simulation is useful for assessing how well the system as a whole will perform its role in a given environment. Systems will typically be analyzed using stochastic discrete event simulations. These are often called Monte Carlo simulations, as random numbers are drawn to represent outcomes from distributions for the input uncertainties. These models are run multiple times and the output probabilities of occurrence are established

by aggregating the simulation runs and using statistical tools for interpretation (Fishman, 1997).

Decision analysis uses a model of the decision makers' preferences and risk attitudes to analyze tradeoffs between alternatives and is often used for strategic decision making.

What is its relationship to SE?

OR techniques frequently support SE by assisting in understanding stakeholder needs and estimating how a proposed system will behave. Decision analysis is used to elicit stakeholder needs and preferences to construct a value model. The value model is used to understand value tradeoffs between different system or system element alternatives. The performance of the anticipated system alternatives may be represented through a simulation. This has been so successful that model-based SE (MBSE) has become an important area (see [Section 4.2.1](#)). Optimization may be applied to improve designs. For example, by minimizing a system element weight while providing required structural strength. Optimization may be used to minimize life cycle costs using inventory theory. Queueing theory may be used to understand system processing times. Forecasting may be used to project forward from historical data. Network flow analysis may be used to improve network capacity.

Blanchard and Fabrycky (2011) discuss the SE morphology for product (system) realization. They note the steps after design synthesis are (1) Estimation and Prediction, and (2) Design Evaluation. Estimation and Prediction often rely heavily on OR. For example, assessing the response times associated with different possible locations for a new firehouse or estimating spare parts requirements as part of a life cycle cost perspective.

Optimization may be applied in the development stage as well as in the production stage. Optimization may be applied at higher levels of the systems hierarchy. For example, a less costly design may support a larger number of employed systems for the same results.

Operations research techniques may be applied to historical and test data. Data analytics may be used to establish system performance as a mathematical function of inputs. Sensitivity analysis may be used to establish acceptable system inputs. Response surface methodology may be used to establish more abstract but tractable models. Forecasting methods may be applied to time series data.

Decision analysis provides the ability to understand decision quality and assess the value or utility delivered under certainty and uncertainty. Decision analysis provides not only the ability to select the best presented alternative but also supports insight in examination of hybrid alternatives, allowing searching for improved designs. It is also transparent and traceable, and so is fully defensible and allows representation of differing preferences of various stakeholders.

How does it impact/is it impacted by SE?

OR and SE practitioners must work closely with each other to accomplish the goals of the project. SE practitioners can perform OR analyses or use OR studies performed by others to make system decisions.

5.4 DIGITAL ENGINEERING

Definitions

The US Defense Acquisition University Glossary (DAU, 2022) defines Digital Engineering as:

An integrated digital approach that uses authoritative sources of systems' data and models as a continuum across disciplines to support life cycle activities from concept through [retirement].

Digital Engineering is crosscutting: it includes all engineering disciplines using well-formed models to execute their processes and communicate a system's design. Digital Engineering emphasizes continuity of models and

their use across the life cycle. Digital Engineering seeks to maximize the use of models and the computer and places emphasis on consistent and rigorous engineering, strong data management practices, and continuous improvement through technologic innovation. It requires a supporting infrastructure and environment and a capable workforce and culture that is committed to working in accordance with process, following methods, and using tool the organization supplies them.

Model-based SE (MBSE) (see [Section 4.2.1](#)) is one of core elements of Digital Engineering. In MBSE and Digital Engineering, a *Digital System Model* is a digital representation of a system. It integrates the authoritative MBSE and other Digital Engineering technical data and associated artifacts, defining all aspects of the system throughout the system life cycle. The Digital System Model is composed of a federated set of models that serve as an authoritative source of truth (ASOT) for the system's design.

Digital Engineering leverages MBSE and the Digital System Model to enable digital threads and digital twins. A *digital twin* is a digital surrogate of the system, incorporating models to emulate the actual system or some of its elements. The digital twin evolves through the life cycle with the mission and definition of the system. A *digital thread* is a set of interconnected, cross-discipline model data that seamlessly expedite the controlled interplay of digital artifacts to inform decision makers throughout a system's life cycle. Digital threads can be used to produce *digital artifacts* that are a combination of authoritative data, information, knowledge, and wisdom addressing stakeholders' unique perspective.

Digital Engineering in Projects

Digital Engineering performed on projects needs to be based around the project and SoI requirements, as well as opportunities and risks that are identified from stakeholders including customers, users, organizational leadership, the infrastructure, decision makers, and project managers. It is important to pay special attention to Digital Engineering contributions and inputs when doing the technical planning across the life cycle.

As each life cycle stage is considered, it is critical to research and document opportunities for future project use, as well as for reuse of models, simulations, and data across the life cycle. For example, models and simulations developed during earlier life cycle stages may be reused for verification and training in later life cycle stages.

Digital Engineering for the Enterprise

Digital Engineering is an approach for projects, but is also important as an enterprise digital transformation initiative (see [Section 5.5](#)). To achieve maximum benefit both project and enterprise level concerns should be considered together. Enterprise level implementation efforts are necessary to position the engineering infrastructure and environment so that a project will be able to perform their engineering activities in a digital manner. Consistent use of the infrastructure and environment on a project, and across projects, will yield increasingly consistent engineering work products. Consistency across work products will lead to great gains in reuse and will enable greater speed through computer automation.

Digital Engineering Ecosystem

The Digital Engineering ecosystem can be vast, crossing engineering domain and organizational boundaries. The Digital Engineering ecosystem should be treated as an SoS and must be developed using SE good practices. The Digital Engineering ecosystem consists of the models, tools, processes, and people/roles that come together to develop the systems the organization cares about. The ecosystem can be broader than this, depending on the scope. For example, some organizations utilize their Digital Engineering ecosystems to support the operation of their systems. Others include cross-project libraries and methods in their Digital Engineering ecosystems, as well as interconnections with external Digital Engineering ecosystems across their supply chain.

Technological Innovation

Technological innovation is an important part of Digital Engineering. It is the catalyst that drives change into the engineering practice through strategic and planned implementation. It is an essential part of a continuous process improvement program that seeks out and injects technology into the systems that are used to develop the SoI. Technological innovation seeks to optimize the use of computer and information technologies to enhance the speed, agility, quality, and precision of all engineering activities that occur through the development life cycle.

5.5 SYSTEMS ENGINEERING TRANSFORMATION

In discussing SE transformation, there are three important transformations to be considered:

- Transforming from no SE to full use of SE,
- The internal SE from traditional to agile methodologies as appropriate, and
- The internal SE from document-based to model-based disciplines.

Considering the first, for an organization to begin implementing SE requires a cultural shift to understand the basics of the system life cycle and implement the life cycle processes described in this handbook in the organization's set of processes (see [Section 2.3.3.1](#)). When performing this type of transformation, organizations will need to prepare for the major cultural shift required for the team members to learn, understand, and perform SE tasks.

The second type of transformation for internal SE from traditional to agile methodologies also requires a cultural shift to understand the basics of agile SE (see [Section 4.2.2](#)) in the organization's set of processes. Agile SE requires tools and capabilities that may also cause the organization to need additional infrastructure and human talent. Every organization will need to determine the best way to integrate agile SE practices and

methodologies, whether as a wholesale change or a gradual change for their SE processes. In practice, the main reasons why are required these two types of transformations are that the system life cycle is either being compressed during the concept and development stages and/or the life cycle is being shifted to the left with early concept task being anticipated with early involvement of SE.

The third transformation is SE's transformation to a model-based discipline supports transdisciplinary digital engineering through model-based activity, advancement, organizational change, and a broad model community engagement (Peterson 2019). This transformation addresses:

- Knowledge representation and immersive technologies,
- Product (System) modeling,
- Model-based SE (MBSE) approaches and methods,
- Virtual prototyping and virtual product integration at scale,
- Foundational theory, principles, and heuristics, and
- MBSE in support of Digital Engineering.

Knowledge representation and immersive virtual reality technologies (characterized by deep absorption or immersion) enable highly efficient and shared human understanding of systems in a virtual environment that spans the full life cycle. Systems modeling forms the product-centric backbone of the digital enterprise which incorporates a model-centric approach to integrate technical, programmatic, business, regulatory, and governance concerns (see [Section 3.2.1](#)). Model-based approaches and sophisticated model-based methods extend beyond product modeling to include systems of systems (SoSs) and enterprise-level modeling and analysis (see [Section 4.2.1](#)). Large scale virtual prototyping and virtual product integration based on integrated models enable significant time-to-market reductions (see [Sections 3.2.2](#) and [2.3.5.8](#)). Foundations of theory, principles, and heuristics allow for a better understanding of increasingly complex systems and decisions in the face of uncertainty (see [Section 1.4](#)). MBSE in support of Digital Engineering is standard practice and is

integrated with other modeling and simulation as well as digital enterprise functions (see [Section 5.4](#)).

5.6 FUTURE OF SE

The primary focus of this handbook is the state-of-the-good-practice in SE. This section highlights some emerging areas, looking toward the future of SE. INCOSE's Systems Engineering Vision (2022) is an excellent resource. In addition, the SE Body of Knowledge (2023) is continually updated to reflect both the state-of-the-practice and the state-of-the-art in SE.

INCOSE's Future of Systems Engineering (FuSE) is a systems community initiative to realize the INCOSE Vision 2035 (2022) and to evolve the instruction, practice, and perception of SE to:

- Position the discipline to leverage new technologies in collaboration with allied fields,
- Enhance SE's ability to solve the emerging challenges, and
- Promote SE as essential for achieving success and delivering value in the engineering of socio-cyber-physical systems and SoSs at scale and subject to non-deterministic influences and effects.

An important aspect of SE is to keep current on emerging trends, technologies, and challenges when considering both the SoI and the SE processes themselves. While performing SE, the practitioner needs to consider advances in computing, communications, software, human systems integration, and algorithms such as augmented intelligence and Machine Learning (ML) for both use in systems and SE. Leveraging these technologies will increase the capabilities of systems and the SE practitioner. By *scale*, we mean the challenges in applying SE from smaller (e.g., miniaturization of electronics) through larger scales (e.g., cloud-based systems with millions of users). With the exponential growth of scale and hyperconnectivity of systems and SoSs, scale is becoming even more important to the discipline of SE.

At an early 2019 FuSE workshop hosted by INCOSE, the terms Artificial Intelligence (AI) for SE and SE for AI were first used to describe the dual transformation envisioned for both the SE and AI disciplines. The “AI4SE” and “SE4AI” labels have quickly become symbols for an upcoming rapid evolutionary phase in the SE community. AI4SE applies AI and ML techniques to improve human-driven SE practices. This goal of “augmented intelligence” includes outcomes such as achieving scale in model construction and efficiency. Enhancing and assisting SE processes, methods, and tools, with tangible impacts on the quality of the engineered system as well as on the cycle time for the various life cycle activities, would be some of the primary focus areas of AI4SE (SEBoK, 2023).

The FuSE roadmap drives this evolution of SE to:

- Be increasingly adaptable, evolvable, and fit for purpose,
- Account for human abilities and needs as an integral system element and human interactions with systems and SoSs,
- Be more responsive in resolving increasingly challenging societal needs, and
- Realize and enhance the INCOSE Vision 2035 (2022) and other visionary statements.

Greater understanding of the inter-coupled technical, economic, social, and environmental systems will provide the basis for significantly increased involvement of SE practitioners in the policy arena. In this expanded role, SE practitioners will also make important contributions to the design of viable systems and transition pathways supporting global sustainability transformation. The scope of SE will widen to recognize and include policy, legal, economic, and environmental specializations.

6

CASE STUDIES

Real-world examples that draw from diverse industries and types of systems are provided throughout this handbook, and in this part, five case studies have been selected to illustrate the diversity of systems to which Systems Engineering (SE) principles and practices can be applied: medical therapy equipment, a bridge, a breach of a cybersecurity system, a redesign of a high-tech medical system for low-tech maintenance, and autonomous vehicles. They represent examples of failed and successful systems.

6.1 CASE 1: RADIATION THERAPY—THE THERAC-25

Background Therac-25, a dual-mode medical linear accelerator (LINAC), was developed by the medical division of the Atomic Energy Commission Limited (AECL) of Canada, starting in 1976. A completely computerized system became commercially available in 1982. This new machine could be built at lower production cost, resulting in lower prices for the customers. However, a series of tragic accidents led to the recommended recall and discontinuation of the system.

The Therac-25 was a medical LINAC, or particle accelerator, capable of increasing the energy of electrically charged atomic particles. LINACs accelerate charged particles by introducing an electric field to produce particle beams (i.e., radiation), which are then focused by magnets. Medical LINACs are used to treat cancer patients by exposing malignant cells to radiation. Since malignant tissues are more sensitive than normal tissues to radiation exposure, a treatment plan can be developed that permits the absorption of an amount of radiation that is fatal to tumors but causes relatively minor damage to surrounding tissue.

Six accidents involving enormous radiation overdoses to patients took place between 1985 and 1987. Tragically, three of these accidents resulted in the death of the patients. This case is ranked in the top ten worst software-related incidents on many lists. Details of the accidents and analysis of the case are available from many sources, including Jacky (1989) and Leveson and Turner (1993).

Approach Therac-25 was a revolutionary design compared to its predecessors, Therac-6 and Therac-20, both with exceptional safety records. It was based on a double-pass concept that allowed a more powerful accelerator to be built into a compact and versatile machine. AECL designed Therac-25 to fully utilize the potential of software control. While Therac-6 and Therac-20 were built as stand-alone machines and could be operated without a computer, Therac-25 depended on a tight integration of software and hardware. In the new, tightly coupled system, AECL used software to monitor the state of the machine and to ensure its proper operations and safety. Previous versions had included independent circuits to monitor the status of the beam as well as hardware interlocks that prevented the machine from delivering radiation doses that were too high or from performing any unsafe operation that could potentially harm the patient. In Therac-25, AECL decided not to duplicate these hardware interlocks since the software already performed status checks and handled all the malfunctions. This meant that the Therac-25 software had far more responsibility for safety than the software in the previous models. If, in the course of treatment, the software detected a minor malfunction, it would pause the treatment. In this case, the procedure could be restarted by pressing a single “proceed” key. Only if a serious malfunction was detected was it required to completely reset the treatment parameters to restart the machine.

The software for Therac-25 was developed from the Therac-20’s software, which was developed from the Therac-6’s software, a brownfield, or legacy, development (see [Section 4.3.2](#)). One programmer, over several years, evolved the Therac-6 software into the Therac-25 software. A stand-alone, real-time operating system was added along with application software written in assembly language and tested as a part of the Therac-25 system operation. In addition, significant adjustments had been made to simplify the

operator interface and minimize data entry, since initial operators complained that it took too long to enter a treatment plan.

At the time of its introduction to market in 1982, Therac-25 was classified as a Class II medical device. Since the Therac-25 software was based on software used in the earlier Therac-20 and Therac-6 models, Therac-25 was approved by the federal Food and Drug Administration under Premarket Equivalency.

Conclusions The errors were introduced in the concept and early development stages, when the decisions were made to create the software for Therac-25 using the modification of existing software from the two prior machines. The consequences of these actions were difficult to assess at the time, because the starting point (software from Therac-6) was a poorly documented product and no one except the original software developer could follow the logic (Leveson and Turner, 1993).

The issues from the Therac case are, unfortunately, still relevant, as evidenced by similar deaths for similar reasons in 2007 upon the introduction of new LINAC-based radiation therapy machines (Bogdanich, 2010).

6.2 CASE 2: JOINING TWO COUNTRIES—THE ØRESUND BRIDGE

Background The Øresund Region is composed of eastern Denmark and southern Sweden and since 2000 has been linked by the Øresund Bridge. The area includes two major cities, Copenhagen and Malmö, has a population of three million, and counts as Europe's eighth largest economic center. One fifth of the total Danish and Swedish Gross National Product (GNP) is produced in the region. The official name of the bridge is translated "the Øresund Connection" to underscore the full integration of the region. For the first time ever, Sweden is joined permanently to the mainland of Europe by a 10-minute drive or train ride. The cost for the entire Øresund Connection construction project was calculated at 30.1 billion DKK (3 billion USD), and the investment is expected to be paid back by 2035.

The Øresund Bridge is the world's largest composite structure, has the longest cable-stayed bridge span in the world carrying motorway and railway traffic, and boasts the highest freestanding pylons. The 7.9 km (5 miles)-long bridge crosses the international navigation route between the Baltic Sea and the North Sea. A cable-stayed high bridge rises 57 m (160 ft) above the surface of the sea, with a main span of 490 m (0.3 miles). Both the main span and the approach bridges are constructed as a two-level composite steel-concrete structure. The upper deck carries a four-lane motorway, and the lower deck carries a two-track railway for both passenger trains and freight trains. The rest of the distance is spanned by the artificial island Peberholm ("Pepper" islet, named to complement the Saltholm islet to the north) and a tunnel on the Danish side that is the longest immersed concrete tunnel in the world. Since completion, Peberholm has become a natural habitat for colonies of rare birds, one of the largest of its kind in Denmark and Sweden.

Nations other than Denmark and Sweden also contributed to this project. Canada provided a floating crane, aptly named Svanen (the swan), to carry prefabricated bridge sections out to the site and place them into position. Forty-nine steel girders for the approach bridges were fabricated in Cádiz, Spain. A specially designed catamaran was built to handle transportation of the foundations for the pylons, which weighed 19,000 tons each.

Approach As noted in the many histories of the bridge, the development stage of the project began with well-defined time, budget, and quality constraints. The design evolved over more than seven years, from start to delivery of final documentation and maintenance manuals. More than 4,000 drawings were produced. The consortium dealt with changes, as necessary, using a combination of technical competence and stakeholder cooperation. Notably, there were no disputes and no significant claims against the owners at the conclusion, and this has been attributed to the spirit of partnership.

What is not often reported is that the success of the development stage is clearly based on the productive, focused, creative effort in the concept stage that began when the royal families of Denmark and Sweden finally agreed in 1990 to move ahead with a bridge project connecting their two countries. That SE effort shaped the approach to the project with well-defined time,